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# The Cradles of Planets: An Overview of Protoplanetary Disk Theory and Observation

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**ABSTRACT**

Protoplanetary disks of gas and dust around young stars are the birthplaces of planets. We review how these disks form during the gravitational collapse of molecular cloud cores and the physical and chemical conditions expected within them, with an emphasis on surface density and temperature profiles and the location of snow lines. We then describe the main observational techniques used to study disks: thermal continuum imaging, molecular spectroscopy, polarimetry, and near-infrared imaging. We close by outlining open questions in disk formation, the limitations of current models, and the observational advances needed to further constrain planet formation.

**Key words.** Protoplanetary disks, Planets and satellites: formation, Stars: formation, Techniques: interferometric, Techniques: polarimetric, Techniques: spectroscopic

## 1. Introduction

Looking at our solar system, it is intriguing how all planets orbit the Sun in the same direction, with similarly low eccentricities, and in approximately the same plane. To explain this configuration, Swedenborg, Kant, and Laplace introduced in the 18th century the nebular hypothesis, suggesting that planets evolved from a rotating disk of gas and dust around the Sun (Hogerheijde 2010). Explaining the origin and composition of this disk was unfeasible with the technology available back then, but subsequent work revealed that protoplanetary disks originate from the gravitational collapse of molecular clouds primarily composed of hydrogen and helium. The essence of the nebular hypothesis is still valid to this day, even generalized to planets orbiting stars beyond our Sun.

The first detection of a circumstellar disk goes back to 1984, when astronomers at the Las Campanas Observatory observed a disk around the star $\beta$ Pictoris. This disk was detected in the infrared and extends out to 1500 AU from the star, with a relatively dust-free region within the inner 100 AU (Smith and Terrile 1984). Since then, more than a thousand protoplanetary disks have been detected, 873 of them in the L1641 and L1647 regions of the Orion A cloud using the Atacama Large Millimeter/submillimeter Array (ALMA) at 225 GHz (Van Terwisga et al. 2022).

Section 2 of this article provides an overview of the mechanisms and conditions that play a role in protoplanetary disk formation. Section 3 discusses the physical conditions within the disks, notably temperature and density profiles. Section 4 elaborates on the techniques used in disk observation, and Section 5 summarizes open challenges.

## 2. Formation of the Protoplanetary Disk

During star formation, a molecular cloud composed primarily of hydrogen and helium begins to collapse under the force of gravity. This happens once the cloud reaches a critical mass and size, at which point gravity becomes too strong for thermal and magnetic pressure to support the cloud against its own weight. In an isothermal, uniform medium of density $\rho_0$, the minimum wavelength for gravitational fragmentation is the Jeans length

$$\lambda_J = \sqrt{\frac{\pi\sigma_{\rm th}^2}{G\rho_0}} = \pi^{1/2} R_G(\rho_0),$$

where $R_G = \sigma_{\rm th}/\sqrt{G\rho_0}$, $\sigma_{\rm th} = \sqrt{P_{\rm th}/\rho_0}$ is the thermal velocity dispersion, and $P_{\rm th}$ is the thermal pressure. The corresponding Jeans mass is

$$M_J \equiv \frac{4\pi}{3}\left(\frac{\lambda_J}{2}\right)^3 \rho_0 = 2.47\, M_{\rm BE},$$

where $M_{\rm BE}$, the Bonnor-Ebert mass, is the maximum mass of a static isothermal cloud that can remain in hydrostatic equilibrium (Bonnar 1956; Ebert 1957):

$$M_{\rm BE} \equiv 1.182\, \frac{\sigma_{\rm th}^4}{\left(G^3 P_{\rm th,0}\right)^{1/2}} = 1.182\, \frac{\sigma_{\rm th}^3}{\left(G^3\rho_0\right)^{1/2}}.$$

The collapse leads to the formation of a solar nebula with a protostar at its center. As the cloud contracts, its rotation rate increases due to the conservation of angular momentum. This accelerated rotation causes the cloud's material to accumulate in a plane perpendicular to the net angular momentum of the original cloud, flattening the cloud

into a thin, rotating disk known as the protoplanetary disk (O'Dell and Beckwith 1997).

Material falling in from the outer regions of the nebula carries high specific angular momentum, so it lands on the disk rather than directly on the protostar. Consequently, disks are expected to grow quickly: the centrifugal radius, inside which infalling material settles onto the disk, grows as $R_c \propto \Omega^2 t^3$, where $t$ is the time since the onset of collapse and $\Omega$ is the rotation rate of the core (Terebey et al. 1984). Observations have shown, however, that the angular momentum of the gas in cores is small. It is characterized by

$$\beta_{\rm rot} \equiv \frac{1}{3}\frac{v_{\rm rot}^2 R}{GM},$$

the ratio of the rotational energy to the gravitational binding energy for a uniformly rotating, constant-density sphere. The median value of $\beta_{\rm rot}$ is around 0.03 (McKee and Ostriker 2007).

Measuring continuum emission at different millimeter wavelengths is commonly used to estimate the dust mass of an emitting region. Using the integrated disk flux $F_\nu$, and assuming optically thin emission, the dust mass of the disk is

$$M_{\rm dust} = \frac{D^2 F_\nu}{\kappa_\nu B_\nu(T_{\rm dust})},$$

where $D$ is the distance to the source, $\kappa_\nu$ is the dust opacity, and $B_\nu(T_{\rm dust})$ is the Planck function evaluated at the dust temperature $T_{\rm dust}$. In dense protostellar envelopes, the dust temperature is usually set to 30 K (Tychoniec et al. 2020).

## 3. Physical conditions in the disk: Temperature, Density profiles

### 3.1. Physical and Chemical Composition

Protoplanetary disks are 99% gas by mass. The remaining mass takes the form of dust: solid particles that permeate the gas and range in size from micrometer-sized grains to centimeter-sized pebbles. These dust particles are the fundamental building blocks of asteroids and planets.

The chemical composition of a protoplanetary disk depends on the cloud from which it forms. The gas is predominantly $H_2$ (molecular hydrogen) and He (helium). It is usually characterized by a mean particle mass

$$\mu = \frac{M_g}{N},$$

where $M_g$ is the total mass of the gas in the disk and $N$ the number of particles in it. $\mu$ is typically around 2.3 $m_p$ (proton masses).

In addition to hydrogen and helium, traces of heavier elements are present in the gas and in the dust. Some of these elements are almost exclusively found in the solid phase, i.e. in the dust, for the temperature and pressure ranges expected in a protoplanetary disk. These elements are dubbed 'refractory' and are often rocky in nature. Refractory matter in the disk varies in chemical composition. A significant amount of it is minerals containing Si (silicon) and O (oxygen), commonly referred to as silicates. In contrast with refractory materials, protoplanetary disks also house elements known as volatiles, which sublimate (transition from solid to gas phase) at temperatures as low as 20 K. Some prime examples are water ($H_2O$), carbon monoxide (CO), and methane ($CH_4$). Because of their low sublimation temperatures, these species are found in the gas in the inner (hotter) part of the disk. Toward the outer edge, farther from the central star where temperatures are lower, volatiles are found as ices (in the solid phase) on the dust. Dust particles collide and grow, much like dust bunnies, into bigger and more massive agglomerates, ultimately making up the planetesimals that will collide to form planets.

Since dust provides the building blocks for planetesimals and protoplanetary cores, planetesimals forming in the inner disk will incorporate only refractory elements, whereas those forming in the outer disk can incorporate volatile material such as water.

Therefore, understanding the temperature distribution at different locations in the disk and over its lifetime is very important for figuring out where and how planets of different compositions can form.

### 3.2. Surface Density

Now that we have a general idea of what a protoplanetary disk is made of, we can start thinking about how this material is spatially distributed throughout the disk.

Given how predominant the gas is, understanding its density and how it varies throughout the disk goes a long way towards drawing a picture of what a planet-forming disk looks like. Rather than thinking in terms of a three-dimensional quantity (a function of three spatial coordinates $x$, $y$, $z$), we often consider the surface mass density, which varies with $r$, the distance from the central star, also referred to as the heliocentric distance.

To do this, we move from Cartesian coordinates ($x$, $y$, $z$) to cylindrical coordinates: the heliocentric distance $r$, an azimuthal angle $\phi$, and a height above the midplane $z$.

The idea behind the surface mass density begins with the assumption that the disk is azimuthally symmetric, i.e. the density does not vary with azimuthal angle $\phi$: $\rho(r, \phi, z) = \rho(r, z)$. This assumption makes sense because the two main forces acting on the gas, its own pressure and the gravitational pull of the central star, both depend only on the distance from the central star, $d = \sqrt{r^2 + z^2}$. As is always the case with nature, observed protoplanetary disks deviate from this assumption, but it is still good enough for our purposes.

We then eliminate the height dependence by integrating over $z$ to obtain the surface density:

$$\Sigma(r) = \int \rho(r, z)\, dz,$$

with dimensions of mass per unit area.

Besides reducing the dimensionality of the problem, there are other reasons why this quantity is useful. Observations of a protoplanetary disk are naturally integrated along lines of sight. This means that observing a face-on disk gives direct information on the surface density $\Sigma$. However, to draw conclusions about the three-dimensional density $\rho$, we have to make assumptions about how the mass is vertically distributed.

One of the seminal papers in planet formation is **?**, which reconstructed the surface density profile of the solar

protoplanetary disk from the masses of the planets in the solar system and their iron content. Weidenschilling found that the surface density of the solar protoplanetary disk scaled with heliocentric distance as $\Sigma \propto r^{-3/2}$. **?** later normalized this profile to about $1700\,\mathrm{g\,cm^{-2}}$ at 1 AU, the minimum needed to build the current solar system. This model was dubbed the Minimum Mass Solar Nebula (MMSN) and is still heavily referenced in the literature.

Despite its prevalent use, the MMSN comes with some strong caveats. It makes many assumptions about planet formation and how the solar nebula evolved from an initial disk into what we see today. One example is the location where the solar system planets formed. The MMSN assumes that all the planets formed from the material available at their current orbital locations, whereas more recent models of planet formation expect significant movement of the planets during their formation (planetary migration).

### 3.3. Temperature Profile

The temperature of a protoplanetary disk has two main components: irradiative heating and viscous heating.

The irradiative component is simply radiative heating from the central star. Any star is characterized by a luminosity $L_*$, the amount of energy it emits as light per unit time. We can assume that a star emits this light equally in all directions, just like a light bulb. The amount of energy an object receives depends on two things: its size and its distance from the source. For instance, if I put a ball in front of a light bulb, the closer I place the ball to the bulb, the more light it receives, which translates to a bigger shadow on the wall. On the other hand, if I place two balls of different sizes at the same distance from the bulb, the bigger ball receives more light, which once again translates to a bigger shadow on the wall. The energy per unit time that a particle of radius $a$ at a distance $r$ receives from the central star is

$$L_{\mathrm{in}} = \frac{\pi a^2}{4\pi r^2}\, L_*,$$

where $L_*$ is the luminosity of the star. The $4\pi r^2$ term is the surface area of a spherical shell centered on the star (the area over which the star's energy is evenly spread), and $\pi a^2$ is the cross-section of the particle. Their ratio is therefore the fraction of the star's light that hits our particle.

If we assume our particle is a perfect blackbody (it fully absorbs the star's radiation and re-emits it), the energy it emits per unit time over its whole surface is

$$L_{\mathrm{out}} = 4\pi a^2 \sigma_{\mathrm{SB}} T^4,$$

where $\sigma_{\mathrm{SB}}$ is the Stefan-Boltzmann constant. This is known as the Stefan-Boltzmann law. Note that the particle absorbs over its cross-section $\pi a^2$ but emits over its full surface area $4\pi a^2$.

If the particle is in thermal equilibrium, the energy flowing in and out of it must balance:

$$L_{\mathrm{in}} = L_{\mathrm{out}}.$$

Substituting the expressions above and rearranging, we get

$$T^4 = \frac{L_*}{16\pi\sigma_{\mathrm{SB}} r^2},$$

which we can solve for the equilibrium temperature of our particle, often noted $T_{\mathrm{eq}}$.

This gives us a first-order idea of how the temperature of the gas in the disk depends on distance from the central star, $T \propto r^{-1/2}$. Many models use this temperature dependence, but it relies on simplifying assumptions about the geometry of a disk. In particular, disks are known to flare at larger radii, becoming thicker the farther out we go. Due to this effect, we expect a temperature dependence closer to $T \propto r^{-3/7}$.

Temperature depends not only on $r$ but also on $z$, the height above the midplane of the disk. This dependence is less well constrained. Models usually compute a midplane temperature $T_m$ and a surface temperature $T_s$, then interpolate between the two for intermediate heights. An example of this is the Type II profile of **?**:

$$T(r,z) = T_s(r) + [T_m(r) - T_s(r)] \cos^2\left(\frac{\pi z}{2 Z_q(r)}\right),$$

where $Z_q(r)$ is the critical height at radius $r$, defined as $Z_q(r) = Z_{q,0}\,(r/r_0)^{1.3}$. This model assumes that for any height $z \geq Z_q$ the temperature is constant.

We care about the temperature because, as mentioned in Sect. 3.1, it determines the physical state of the material. Each species can be either in the gas or in the dust, and the pressure and temperature conditions under which it sublimates from the dust or condenses from the gas are important. Using the temperature profile, we can locate where these transitions happen for a given species. We call this location a snow line.

For instance, we can locate the water snow line around the Sun by solving

$$r = \left(\frac{L_\odot}{16\pi\sigma_{\mathrm{SB}} T^4}\right)^{1/2},$$

where we plug in for $T$ the temperature we assume for water freeze-out.

## 4. Disk Observations

Over the past decade, advances in astronomical instrumentation have transformed our understanding of protoplanetary disks, exposing their fine-scale structures, dynamic processes, and chemical complexity. By combining observations across multiple wavelengths, from millimeter-wave interferometry to infrared spectroscopy, we can now trace the evolutionary pathways through which planets emerge and develop in these circumstellar environments. *Interferometry* combines signals from multiple telescopes to achieve angular resolutions far exceeding those of single dishes, enabling us to resolve fine disk structures at scales of a few astronomical units. *Spectroscopy*, on the other hand, disperses light into its constituent wavelengths, allowing us to identify and map the distribution of specific molecules and trace the chemical conditions within disks.

This section explores the main discoveries in protoplanetary disk research, focusing on four observational techniques: thermal continuum imaging, molecular spectroscopy, polarimetry, and near-infrared imaging.

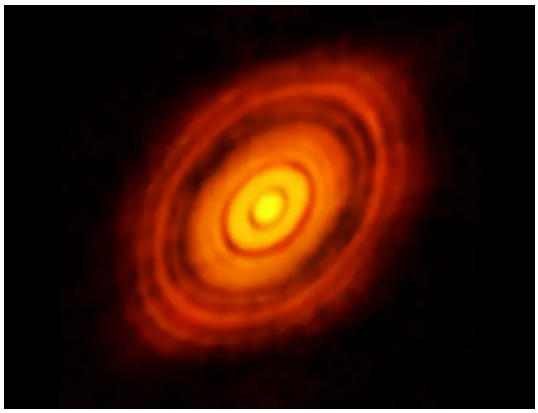


**Fig. 1.** ALMA image of the HL Tau protoplanetary disk, showing rings and gaps that may trace planet formation, though other mechanisms such as pressure bumps and disk instabilities can also produce such structures. (Credit: ALMA (ESO/NAOJ/NRAO))

### 4.1. Thermal Continuum Imaging

Thermal continuum imaging observes infrared to submillimeter emission from dust grains heated by the star or by disk processes. It maps disk features like rings, gaps, and spirals, revealing the dust distribution and temperature (Andrews 2020). ALMA provides sharp images with 2–5 AU resolution. The DSHARP survey (Andrews et al. 2018) found substructures in nearly all observed disks, suggesting that planet formation proceeds quickly.

Key ALMA observations include:

- **HL Tau**: ALMA's images of HL Tau (Fig. 1) show rings and gaps at a resolution of about 35 milliarcseconds, or roughly 5 AU at the distance of HL Tau (ALMA Partnership et al. 2015). These features may be formed by planets, or by dust trapped in pressure bumps (local maxima in gas pressure where the outward pressure gradient force balances inward drag, causing dust particles to accumulate and collide to form larger agglomerates, as discussed in Sect. 3.1). Either way, they may indicate multiple planets shaping the disk.
- **Disk substructures**: Gaps at 1, 22, and 85 au in several disks, noted by Huang et al. (2018), suggest planets or disk instabilities clearing material. These gaps point to planets with masses ranging from sub-Jovian (less than Jupiter's mass) to several Jupiter masses.

To understand the chemical environment of these disks, we turn to molecular spectroscopy.

### 4.2. Molecular Spectroscopy

Molecular spectroscopy studies the chemical makeup of protoplanetary disks by analyzing gas emission lines. During the gas-rich phase of planet formation (the early stage when the disk still retains substantial molecular hydrogen and helium, typically lasting a few million years), disks host complex chemistry driven by high density, warm temperatures, and strong UV and X-ray radiation.

The James Webb Space Telescope (JWST) observes this chemistry through infrared emission from molecules like $H_2O$, CO, $CO_2$, OH, and organic compounds. Some of these species, such as $H_2O$, are very hard to observe from the ground because Earth's atmosphere absorbs at the same wavelengths, which makes a space telescope essential.

JWST's NIRSpec and MIRI instruments detect molecular bands at temperatures of 100–1000 K, with a resolving power of $\lambda/\Delta\lambda \approx 2700$ (where $\lambda$ is the observed wavelength and $\Delta\lambda$ is the smallest wavelength difference the instrument can distinguish), allowing them to identify faint isotopic emission. Their high sensitivity and integral field units map molecular emission in nearby disks and even in distant regions like the Magellanic Clouds (Space Telescope Science Institute 2025).

Next, we explore how polarimetry improves our view of disk structures.

### 4.3. Polarimetry

Polarimetry helps image faint protoplanetary disks around bright host stars by detecting polarized light scattered by dust grains within the disk (De Boer et al. 2020). Unlike the star's unpolarized light, light scattered by disks or planetary atmospheres is polarized (meaning the light waves oscillate preferentially in specific directions rather than randomly). This allows instruments like SPHERE on ESO's Very Large Telescope to suppress stellar glare and reveal disk structures. Observations by SPHERE in 2016 showed that disks are not simple and flat but have spirals, rings, gaps, and shadows (ESO 2016). A 2020 study also found waves and twists in a disk, suggesting the gravitational influence of a forming planet (ESO 2020).

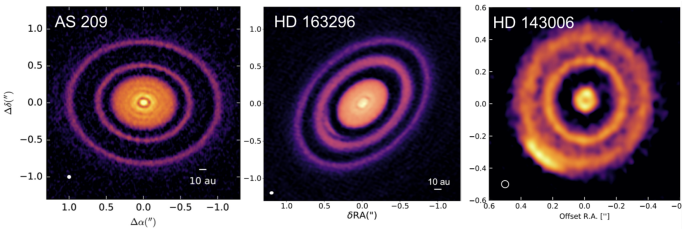



**Fig. 2.** Three protoplanetary disks from the DSHARP survey, observed at millimeter wavelengths and tracing millimeter-sized dust grains. Prominent substructures are visible, such as dark annular gaps separating bright dust rings.

Finally, near-infrared imaging complements these techniques by probing disk surfaces.

### 4.4. Near-Infrared Imaging

Near-infrared (NIR, 1 µm to 5 µm) imaging observes light scattered by small dust grains in the upper layers of protoplanetary disks, uncovering their structure and evolution (Fukagawa et al. 2013). Using techniques such as Polarization Differential Imaging (PDI) and adaptive optics (AO), NIR imaging achieves sharp, high-contrast images with resolutions of approximately 0.05 to 0.1 arcseconds, equivalent to 7 to 14 au in nearby star-forming regions like Taurus. PDI separates the polarized light scattered by dust from the star's unpolarized light, reducing stellar interference. For instance, the SEEDS survey, using the Subaru Telescope's AO188 system and HiCIAO camera, observed disks around young stars and identified features such as spirals and gaps that suggest planet formation (Tamura 2009). NIR imaging probes disk regions from about 0.1 au, just

outside the dust-sublimation zone, to around 100 au, complementing ALMA's observations of larger grains deeper in the disk (Fig. 2). Combining NIR and ALMA data improves our understanding of disk dynamics and planet formation (Fukagawa et al. 2013).

## 5. Conclusion

This article covered the origin of protoplanetary disks and their formation alongside the star during cloud collapse. We discussed, in broad strokes, the expected chemical make-up and physical conditions of these disks, largely based on planet formation in our own solar system. We then described the observational work that allows us to test our planet formation theories against extrasolar systems. Only by gathering more data and further refining our theories will we be able to understand the processes that govern planet formation and explain the current state of our solar system. Using theory and observation in tandem will shed light on how Earth and our neighboring planets formed. This can provide critical information on the origin of life as we know it and the possibility of finding it elsewhere.

However, drawing conclusions with certainty and making strong, evidence-backed claims comes with many challenges. This article highlights some of them:

1. Gaps in our understanding of how a disk forms. While we have a good idea of the conditions needed for star and disk formation to happen, our theory of the details of cloud collapse and material infall remains uncertain.
2. Our theory relies on computational models that we compare against observational data from our own solar system and beyond. An issue inherent to this approach is that our models are ultimately limited by the assumptions that go into them. This creates a delicate balance: we need to make many assumptions to approach the problems at hand, but the resulting model is only as strong as its weakest assumption.
3. Our observational capabilities have made huge strides over the past couple of decades. Nonetheless, the data we have are still technologically limited, and observations at higher spatial and spectral resolution may be required to push our knowledge of planet formation further.

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